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Prediction of stroke-associated hospital-acquired pneumonia: Machine learning approach

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ABSTRACT

Background: Stroke-associated Hospital Acquired Pneumonia (HAP) significantly impacts patient outcomes. This study explores the utility of machine learning models in predicting HAP in stroke patients, leveraging national registry data and SHapley Additive exPlanations (SHAP) analysis to identify key predictive factors.
Methods: We collected data from a national stroke registry covering January 2014 to July 2022, including 9,840 patients diagnosed with ischemic and hemorrhagic strokes. Five machine learning models were trained and evaluated: XGBoost, Random Forest, Support Vector Machine (SVM), Logistic Regression, and Artificial Neural Network (ANN). Performance was assessed using accuracy, precision, recall, F1-score, AUC, log loss, and Brier score. SHAP analysis was conducted to interpret model outputs.
Results: The ANN model demonstrated superior performance, with an F1-score of 0.86 and an AUC of 0.94. SHAP analysis identified key predictors: stroke severity, admission location, Glasgow Coma score (GCS), systolic and diastolic blood pressure at admission, ethnicity, stroke type, mode of arrival, and age. Patients with higher stroke severity, dysphagia, and those arriving by ambulance were at increased risk for HAP.
Conclusion: This study enhances our understanding of early predictive factors for HAP in stroke patients and underlines the potential of machine learning to improve clinical decision-making and personalized care.

Background

Stroke remains a critical global health concern, ranking as the second-leading cause of death and the third-leading cause of death and disability combined, with mortality rates rising particularly in developing nations.[1] Annually, 13.7 million new stroke cases are reported, leading to approximately 5.5 million deaths.[2] The outcome of a stroke is significantly influenced by post-stroke complications, including mobility-related falls, urinary tract infections (UTIs), pneumonia, and psychological conditions.[3,4] Among these, infections are prevalent in the acute phase of stroke, with stroke-associated pneumonia affecting 5–44 % of stroke patients and serving as a major independent risk factor for poor outcomes and increased mortality.[5–7]

Stroke patients face an elevated risk of hospital-acquired pneumonia (HAP) due to factors such as stroke-induced immunodepression, the severity of the stroke, advanced age, ICU admission, pre-stroke disability, and dysphagia.[8–10] HAP, commonly caused by pathogens such as Enterobacteriaceae, Staphylococcus aureus, and Pseudomonas aeruginosa,[11] is often linked to aspiration and impaired reflexes following a stroke.[12,13] It contributes to poorer functional outcomes, increased mortality within the first-year post-stroke, and prolonged hospital stays.[14,15] Preventing pneumonia is, therefore, essential for improving patient outcomes. Recent predictive methods, including the A2DS2 score combined with fasting hyperglycemia,

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neutrophil-to-lymphocyte ratio (NLR), and biomarkers like transthyretin levels and neutrophil percentage to albumin ratio (NPAR), have demonstrated effectiveness in early HAP prediction.[7,16–18]

Advancements in machine learning (ML) and the widespread adoption of electronic health records (EHR) have enhanced the development of predictive models for healthcare applications, including store management. ML models, such as the XGBoost model, have proven particularly efficient in predicting HAP among ischemic stroke patients. Building on these advancements, this study seeks to develop a robust ML model for the early prediction of HAP in stroke patients, addressing the urgent need for preventive measures and improved clinical outcomes.[19–22]

Driven by advancements in machine learning models in healthcare and the need for early prediction and prevention of HAP in stroke patients, our study aims to develop a machine learning model to predict HAP early on.

Methodology

Methods

Our methodology employs a systematic approach to leveraging machine learning techniques to analyze data from a stroke registry. Initially, we collected comprehensive data from the registry, ensuring the inclusion of all relevant variables. The dataset was then divided to train multiple machine-learning models. These models were evaluated, and the best-performing model was selected for further analysis and potential deployment. Fig. 1 illustrates the sequential steps of our machine-learning prediction system.

Ethical approval

The Institutional Research Board (IRB) at Hamad Medical Corporation in Qatar approved this study under reference number MRC-01-22-594.

Study population

Data was collected from the national stroke registry at Hamad General Hospital (HGH) at Hamad Medical Corporation, Qatar, from January 2014 to July 2022. This dataset includes all individuals aged 18 and older admitted to HGH with a primary diagnosis of stroke. Since the registry's inception in Qatar up to July 2022, 15,859 patients have received specialized stroke care at the hospital. This total encompasses patients diagnosed with ischemic and hemorrhagic strokes, transient ischemic attacks (TIAs), and stroke mimics. However, our study focuses explicitly on 9,840 patients diagnosed with ischemic and hemorrhagic strokes.

Baseline variables

The collected data encompassed a wide range of patient information, including demographics, hemodynamic measurements at admission (heart rate and blood pressure), stroke risk factors, pre-existing medical conditions, admission locations, and hospitalization outcomes such as hospital-acquired infections, such as pneumonia. Stroke severity at admission was assessed using the National Institute of Health Stroke Scale (NIHSS) scores.[23,24]

Patients were categorized into five specific ethnic groups based on their reported nationality: Qatari, Middle East and North Africa (MENA) region, South Asia region, Southeast Asia region (as defined by the United Nations geo-scheme), and an "other" category for all other nationalities.[25–27] The distinct classification for Qatari patients was implemented to facilitate meaningful comparisons, considering the unique demographic composition of the country, where expatriates constitute a significant portion of the population.[28,29] This classification method has been consistently applied in previous research on stroke in Qatar.[30–32]

All relevant risk factors, including pre-existing medical conditions and smoking history, were documented during the patient's hospital stay and verified by stroke registry personnel through electronic medical records. A total of 28 variables were used to predict hospital-acquired pneumonia associated with stroke. The details are presented in Table 1.

Outcome measures

This study focused on hospital-acquired pneumonia (HAP) in stroke patients, which was simplified into a binary variable: diagnosed HAP was coded as 1, and non-diagnosed HAP was coded as 0.

Inclusion/exclusion criteria

This study included all adult patients aged 18 years or older diagnosed with stroke. From the initial cohort of 15,859 patients, those diagnosed with either ischemic or hemorrhagic stroke totaled 9,840, comprising 8,183 ischemic stroke cases and 1,657 hemorrhagic stroke

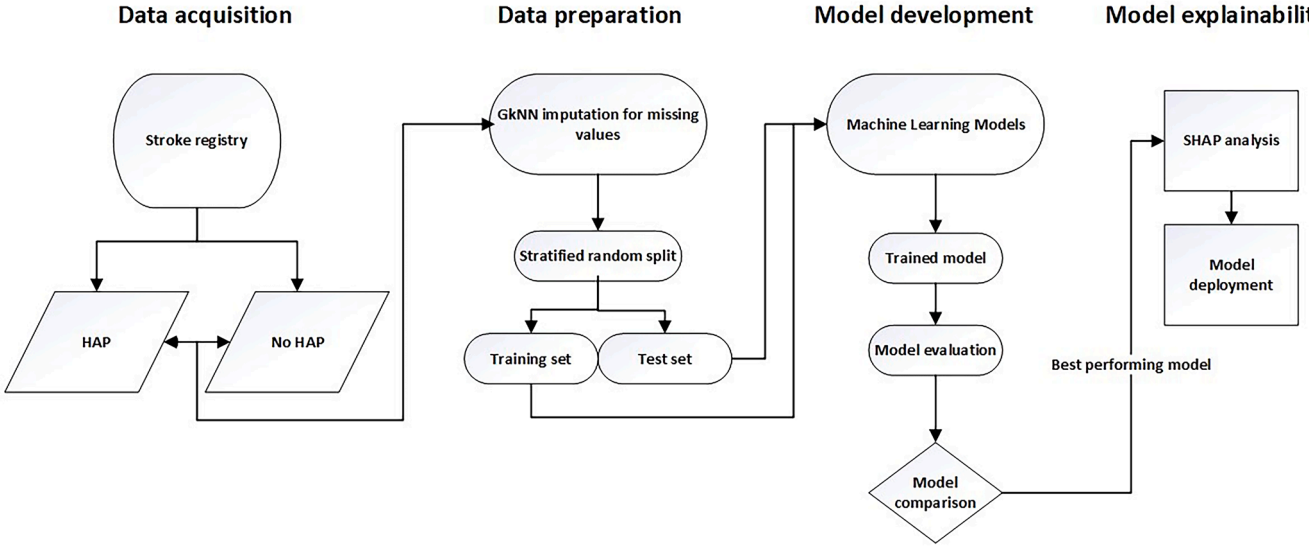


Fig. 1. Flow diagram for the proposed ischemic mortality prediction system.

Table 1
Patients' characteristics.

Variable	Feature	No HAP	HAP	Total
Age (year)	< mean (54.2)	5015	299	5314
(Missing % = 0 %)	≥ mean (54.2)	4243	283	4526
	Mean ± SD (54.2 ± 13.6)-IQR 19			
Sex	1: Male	7450	475	7925
(Missing % = 0 %)	2: Female	1808	107	1915
Ethnicity	1: Qatari	1562	99	1661
(Missing % = 0 %)	2: MENA	1625	105	1730
	3: South Asian	4783	286	5069
	4: South-East Asian	888	64	952
	5: Other	400	28	428
Mode of Arrival	1: Ambulance	6846	504	7350
(Missing % = 0 %)	2: Other	2412	78	2490
Stroke severity at admission	< mean 6.2	6619	91	6710
(NIHSS)	≥ mean 6.2	2639	491	3130
(Missing % = 0 %)	Mean ± SD (6.2 ± 6)-IQR 5			
Onset of symptoms to hospital	1: ≤ 3 h	2778	279	3057
arrival	2: 3-6 h	1154	65	1219
(Missing % = 0 %)	3: > 6 h	5326	238	5564
Glasgow Coma Scale at admission	< mean (14)	1244	287	1531
(GCS)	≥ mean (14)	5951	135	6086
(Missing % = 22.6 %)	Mean ± SD (14 ± 2.4)-IQR 0			
Random Blood Sugar at admission	< mean (9.3)	5930	335	6265
(RBS) mmol/l	≥ mean (9.3)	3131	237	3368
(Missing % = 2.1 %)	Mean ± SD (9.3 ± 4.7)-IQR 5.2			
Heart rate at admission (bpm)	< mean (82.8)	4871	257	5128
(Missing % = 1.6 %)	≥ mean (82.8)	4252	307	4559
	Mean±SD (82.8±15.4)-IQR 19			
Systolic blood pressure at	<mean (160.2)	5168	280	5448
admission (mmHg)	≥mean (160.2)	4054	298	4352
(Missing%= 0.4 %)	Mean±SD (160.2±33)-IQR 44			
Diastolic blood pressure at	<mean (94)	5182	286	5468
admission (mmHg)	≥mean (94)	4033	290	4323
(Missing%= 0.5 %)	Mean ± SD (94 ± 21.2)-IQR 27			
Pre-stroke functional status -	< mean (0.42)	8053	478	8531
modified Rankin Score (mRS)	≥ mean (0.42)	1205	104	1309
(Missing % = 0 %)	Mean ± SD (0.42 ± 1.1)-IQR 0			
Body Mass Index (BMI)	< mean (27.7)	4735	275	5010
(Missing % = 10 %)	≥ mean (27.7)	3626	217	3843
	Mean ± SD (27.7 ± 5.1)-IQR 5.9			
Diabetes Mellitus (DM)	0: No	4409	289	4698
(Missing % = 0 %)	1: Yes	4849	293	5142
Hypertension (HTN)	0: No	2472	156	2628
(Missing % = 0 %)	1: Yes	6786	426	7212
Dyslipidemia	0: No	5237	405	5642
(Missing % = 0 %)	1: Yes	4021	177	4198
Prior stroke	0: No	8235	519	8754
(Missing % = 0 %)	1: Yes	1023	63	1086
Atrial Fibrillation (AF)	0: No	8814	529	9343
(Missing % = 0 %)	1: Yes	444	53	497
Coronary Artery Disease (CAD)	0: No	8234	500	8734
(Missing % = 0 %)	1: Yes	1024	82	1106
Dysphagia	0: No	9129	553	9682
(Missing % = 0 %)	1: Yes	129	29	158
Dysarthria	0: No	6486	462	6948
(Missing % = 0 %)	1: Yes	2772	120	2892
Loss of conciseness (LOC)	0: No	8669	379	9048
(Missing % = 0 %)	1: Yes	589	203	792
Seizure	0: No	9079	553	9632
(Missing % = 0 %)	1: Yes	179	29	208
Tobacco use	0: No	7212	513	7725
(Missing % = 0 %)	1: Yes	2046	69	2115
Thrombolysis	0: No	8417	518	8935
(Missing % = 0 %)	1: Yes	841	64	905
Admission location	1: Stroke Unit	4392	152	4544
(Missing % = 0 %)	2: ICU	818	341	1159
	3: Other	4048	89	4137
Critical care out of ICU	0: No	8948	498	9446
(Missing % = 0 %)	1: Yes	310	84	394
Stroke diagnosis	1: Ischemic	7859	324	8183
(Missing % = 0 %)	2: Hemorrhagic	1399	258	1657
Hospital Acquired Pneumonia	0: No HAP	9258	582	9840
(HAP)	1: HAP			
(Missing % = 0 %)				

cases. The original cohort excluded 1,410 cases of transient ischemic attacks (TIAs) and 4,609 cases of stroke mimics, resulting in a total of 6,019 excluded cases.

Handling missing data and class imbalance

To address missing data values, the Gray K-Nearest Neighbor (GkNN) strategy was employed for imputation. This method treats imputed instances as observed data, which are then combined with complete instances for subsequent rounds of imputation. The GkNN algorithm effectively handles heterogeneous datasets, encompassing numerical and categorical attributes.[33]

The data indicated that the Glasgow Coma Score (GCS) had the highest rate of missing values at 22.6 %, followed by Body Mass Index (BMI) at 10 %, Random Blood Sugar (RBS) at 2 %, heart rate (HR) at 1.6 %, and systolic and diastolic blood pressure (SBP and DBP) at <1 %. All other variables were fully complete. Table 1 presents the percentages of missing values for each variable. The dataset revealed a low HAP rate of 6 %, highlighting a significant class imbalance between the HAP and the No-HAP cohorts. To address this, random under-sampling (RUS) was applied to the minority class, balancing the dataset by randomly removing instances from the majority class until its size matched that of the minority class. This approach effectively mitigated class imbalance issues for the analyses.[34] Additionally, to enhance the performance and efficiency of the models, feature scaling through data normalization was conducted before training the machine learning models.

Model training and evaluation

The dataset was divided into two subsets: a training set (80 %) and a validation set (20 %) using stratified random sampling. The training and testing process included 5-fold cross-validation. Five different machine learning models were trained: extreme gradient boosting (XGBoost), random forest (RF), support vector machine (SVM), logistic regression (LR), and multilayer perceptron artificial neural network (ANN).

To evaluate the performance of these models, we utilized multiple classification metrics, including accuracy, precision, recall, F1-score, area under the receiver operating characteristic curve (AUC), log loss, and Brier score.[35–39] These metrics provide comprehensive insights into the models' ability to accurately classify positive and negative instances while accounting for class imbalance. The model with the highest F1-score will be selected as the primary model for subsequent external and temporal validation.

Additionally, we employed the SHAP (SHapley Additive exPlanations) library to interpret the predictions of the machine learning models.[40] This tool generates individual-level feature importance scores, known as SHAP values, which quantify the contribution of each feature to a specific prediction outcome.

Results

Baseline characteristics

Table 1 presents the characteristics of the study population concerning HAP. In this cohort, 6 % of patients developed HAP. Most of the study population were male, accounting for 81 % of the cohort. The mean age was 54.2±13.6 years. Additionally, 51.5 % of the patients who sought care in the HGH stroke service were of South Asian ethnicity.

Model evaluation

Five distinct models were trained for separate predictions, demonstrating satisfactory predictive power. The ANN model exhibited an impressive F1-Score of 0.86, indicating a strong balance between precision and recall. It also achieved excellent discrimination power, as evidenced by an AUC of 0.94, and demonstrated high predictive

accuracy with a low log loss of 0.13 and a Brier score of 0.12.

Based on these evaluation metrics, we selected the ANN model for further SHAP analysis. Table 2 details the performance metrics of all models, and Fig. 2 illustrates the discrimination power (AUC) for each trained model.

Model explanation- SHAP analysis

The SHAP analysis provided invaluable insights into our predictive model for HAP, identifying the most influential factors in descending order of importance: severity at admission (measured by the NIHSS), admission location, GCS score at admission, systolic blood pressure (SBP) at admission, onset of symptoms, diastolic blood pressure (DBP) at admission, Dysphagia, diagnosis (ischemic vs. hemorrhagic stroke), mode of arrival (MOA), and age. Fig. 3 presents visual representations of these findings.

It is important to note that some factors contribute to the model's performance through association rather than causation. This distinction will be thoroughly examined in the discussion section. Nevertheless, the key variables driving the model's performance are those captured early at admission. Early identification of these critical factors will likely enhance the implementation of preventive measures.

Discussion

This study evaluated the effectiveness of five machine learning models in predicting hospital-acquired pneumonia (HAP) in stroke patients, utilizing SHAP analysis to identify key predictors. The artificial neural network (ANN) demonstrated superior performance among the tested models. The SHAP analysis revealed that certain variables, such as stroke severity and the Glasgow Coma Scale (GCS) score, were strong predictors. In contrast, variables with low SHAP values contributed minimally to the model's predictive accuracy. These critical predictors, assessed upon admission, could help clinicians predict HAP early and personalize care strategies.[41]

Consistent with prior research, stroke severity, measured by the NIHSS, was the most significant predictor of HAP.[10,42] More severe strokes are associated with prolonged hospitalization and greater risks of aspiration and dysphagia, which increase HAP risk.[43] In our study, patients who developed HAP had a significantly higher NIHSS score than those who did not (15.4 vs. 5.8), with HAP risk increasing 13-fold for patients with severity scores above the mean of 6.2 ($p < 0.05$).

The admission location was also a key factor in HAP occurrence.[44,45] Patients admitted to critical care units had a higher incidence of HAP (29.4 %) compared to those in stroke units (3.3 %) and other wards (2.2 %; $p < 0.05$). However, despite the specialized care in stroke units, these patients had a higher incidence of HAP than those in general wards, likely due to the higher severity of stroke at presentation (mean NIHSS score of 6.4 vs. 3.6, $p < 0.05$).

The GCS at admission was another significant predictor of HAP. Patients with lower GCS scores had a higher incidence of HAP (18.7 % vs. 2.2 % for those with a GCS > 14, $p < 0.05$). This increased risk is likely linked to the higher likelihood of aspiration in patients with reduced consciousness post-stroke.[46–48]

Blood pressure at admission also predicted HAP development. Elevated systolic (SBP) and diastolic blood pressure (DBP) were associated with a higher risk of HAP.[32,49] Patients with SBP > 160 mmHg had a 1.5 times higher risk of developing HAP, and those with DBP > 94 mmHg had a 1.3 times higher risk ($p < 0.05$). Additionally, patients with hemorrhagic strokes had significantly higher SBP and DBP than those with ischemic strokes (180.5 mmHg vs. 156 mmHg for SBP and 106.7 mmHg vs. 90 mmHg for DBP, $p < 0.05$).

The time from stroke onset to hospital arrival also influenced HAP risk.[50–53] Patients who arrived within three hours had a higher incidence of HAP (9.1 % vs. 5.3 % for those arriving 3-6 h post-onset, $p < 0.05$). This group had more severe strokes, as reflected by a higher mean NIHSS score of 9 compared to 6 and 5 for the other time intervals ($p < 0.05$). The mode of arrival, particularly by ambulance, was linked to an increased risk of HAP, likely due to the higher severity of cases transported by ambulance (mean NIHSS score of 7.2 vs. 4 for other modes of transport, $p < 0.05$).

Dysphagia was identified as a significant risk factor for HAP, with patients with dysphagia having a threefold increased risk of developing HAP compared to those without.[43,54] In our cohort, 18.4 % of patients with dysphagia developed HAP, compared to only 5.7 % of those without ($p < 0.05$).

Stroke type also predicted HAP risk. Patients with hemorrhagic strokes were five times more likely to develop HAP than those with ischemic strokes (15.6 % vs. 4 %, $p < 0.05$). This higher risk may be attributed to the greater severity of hemorrhagic strokes and associated complications, such as increased intracranial pressure and immune suppression, which increase the risk of infection.[31,55]

Finally, advanced age has been consistently correlated with increased mortality after stroke.[56–58] In our study, patients who developed HAP were significantly older than those who did not (55 vs. 54 years, $p < 0.05$), which may further complicate their stroke recovery.

Overall, this study highlights several significant predictors of HAP, including stroke severity, GCS, blood pressure, dysphagia, and stroke type. Machine learning models, particularly ANN, can effectively identify these factors, aiding in early prediction and personalized care to mitigate HAP risk in stroke patients.

Limitations

Although this study uses national registry data encompassing all patients who received specialized stroke management in the country over a decade, several limitations should be considered when interpreting the findings.

First, the dataset was exclusively sourced from the stroke registry of a single institution, which may limit the generalizability of the results to broader and more diverse healthcare settings and patient demographics. The lack of specific data types, such as imaging findings, comprehensive laboratory test results, and time-to-event data, may have excluded essential factors influencing the prediction of HAP in stroke patients.

The study's retrospective design has inherent biases and limitations in capturing the nuances of clinical practice as it unfolds. Historical data may not reflect the latest advancements in treatment modalities or shifts in patient profiles, potentially affecting the model's relevance to current clinical scenarios.

Class imbalance, with a relatively low number of HAP instances, posed additional challenges in model training and evaluation. Although undersampling techniques addressed this, the potential for model performance bias must be considered.

The model demonstrated high accuracy and recall, yet the inherent complexity of machine learning approaches, particularly ANN, may hinder their straightforward interpretation and practical application in clinical environments. Furthermore, while SHAP analysis identified significant predictive factors, it did not establish causative links between these variables and HAP in stroke patients. In summary, while the study enhances our understanding of factors predictive of HAP in stroke patients, its findings should be interpreted with these limitations in mind.

Table 2
Models' performance metrics.

Model	Accuracy	Recall	Precision	F1 Score	Log Loss	Brier Score	AUC
ANN	0.89	0.85	0.87	0.86	0.13	0.12	0.94
SVM	0.81	0.80	0.82	0.80	0.47	0.15	0.86
RF	0.79	0.78	0.79	0.78	0.46	0.15	0.87
XGBoost	0.80	0.79	0.80	0.79	0.62	0.16	0.86
LR	0.81	0.79	0.82	0.80	0.47	0.15	0.81

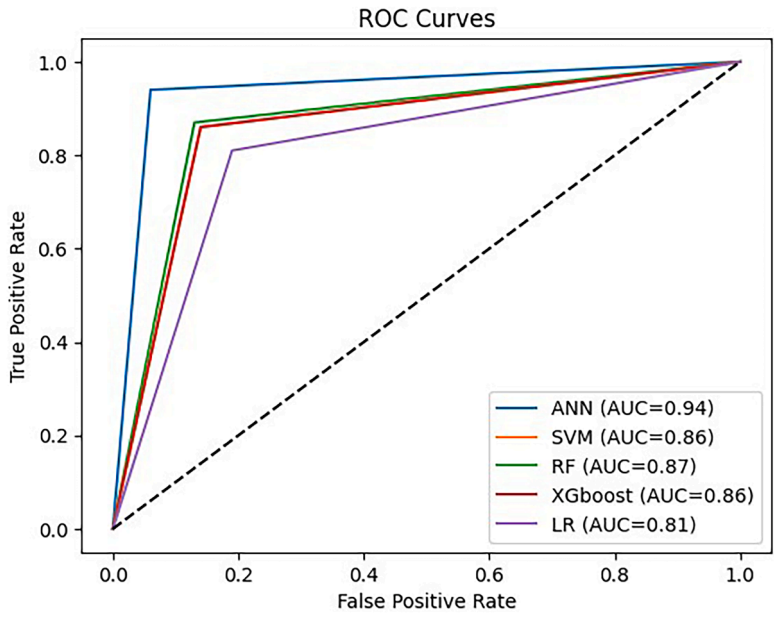


Fig. 2. Area Under the Curve for the trained models.

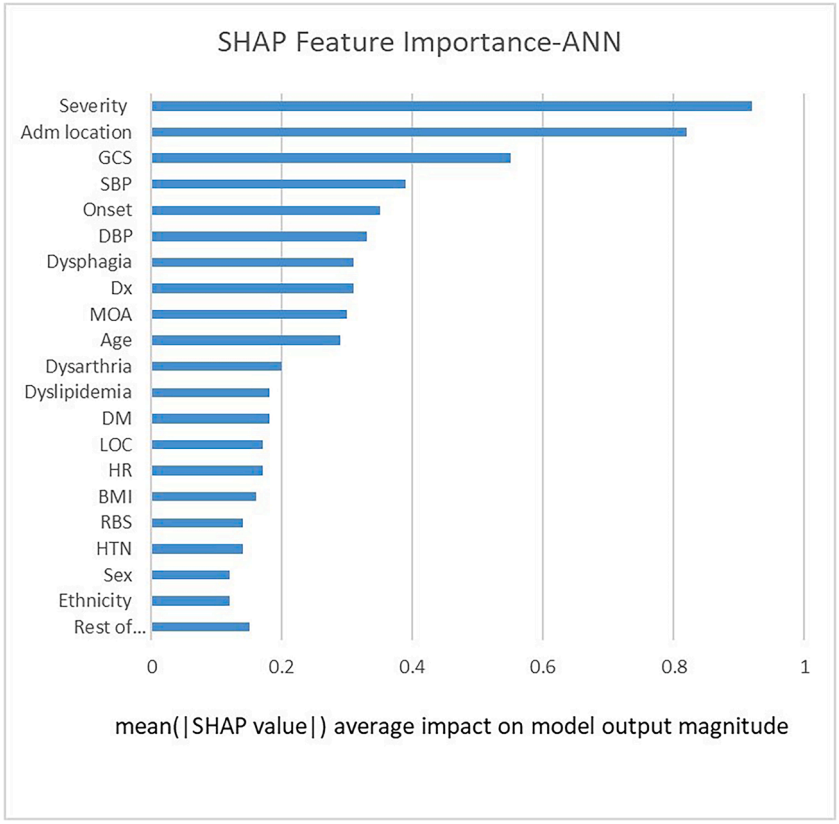


Fig. 3. SHAP feature importance for the ANN- (Mean SHAP value).

Conclusion

This study highlights the effectiveness of machine learning models, particularly the ANN model, in predicting HAP in stroke patients. Identifying key factors such as stroke severity and admission metrics provides valuable insights for early prevention and personalized care. These findings align with existing research while offering specific guidance for improving outcomes. Despite limitations, the study underscores the potential of machine learning in enhancing clinical decision-making. Future research should focus on diverse datasets and prospective designs to validate and expand these findings, advancing predictive accuracy and clinical impact.

Statement of ethics

The IRB at the Medical Research Center of Hamad Medical

Corporation approved this project (MRC-01-22-594). The research was carried out in compliance with the ethical principles outlined in the Helsinki Declaration of 1964 and its subsequent modifications, as well as related ethical norms. The IRB waived the need for informed consent at the Medical Research Center of Hamad Medical Corporation due to the retrospective nature of the study.

Consent for publication

Not applicable.

CRediT authorship contribution statement

Ahmad A. Abujaber: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Said Yaseen:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Abdulqadir J. Nashwan:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Naveed Akhtar:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Yahia Imam:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request and subject to appropriate ethical approvals.

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